

GPG Asymmetric Encryption Documentation (RHEL)

1. Install GPG on RHEL

Update the system and install GnuPG:

```
sudo yum update -y  
sudo yum install gnupg2 -y
```

Verify the installation:

```
gpg --version
```

2. Scenario: User1 & User2 Secure File Sharing

We have two users:

- User1 → Wants to receive a secure file
- User2 → Wants to send a secure file to User1

Step 1: User1 generates a key pair

On User1's machine:

```
gpg --gen-key
```

This creates a Public Key (for sharing) and a Private Key (kept secret).

Step 2: User1 exports the public key

```
gpg --armor --export user1@example.com > user1_pubkey.asc
```

The file user1_pubkey.asc contains User1's public key, which should be sent to User2.

Step 3: User2 imports User1's public key

On User2's machine:

```
gpg --import user1_pubkey.asc
```

Step 4: User2 encrypts the file

Suppose User2 wants to send secret.txt:

```
gpg --encrypt --recipient user1@example.com secret.txt
```

This generates an encrypted file: secret.txt.gpg

Step 5: User2 sends the encrypted file to User1

User2 transfers secret.txt.gpg to User1 (via email, SCP, etc.).

Step 6: User1 decrypts the file

On User1's machine:

```
gpg --decrypt secret.txt.gpg > secret_decrypted.txt
```

User1 will be asked for their passphrase. The decrypted file will be saved as secret_decrypted.txt.

Summary of Workflow

1. User1 creates a key pair.
2. User1 shares their public key with User2.
3. User2 imports the public key.
4. User2 encrypts the file using User1's public key.
5. User2 sends the encrypted file to User1.
6. User1 decrypts the file using their private key.

This ensures:

- Confidentiality → Only User1 can read the file.
- Security → Even if intercepted, the file cannot be decrypted without User1's private key.